# SUBSEA MATTRESS VERIFICATION REPORT: CR-A1_M15 (#A046)

Project: Tung-Hsiao Power Plant 2nd Stage Renewal Subsea Gas Pipeline Project
Client: Taiwan Power Company (TPC) | Contractor: NMDC Energy | Vessel: DP2 MMA Prestige
Crossing Array: CR-A1_L1 | Location ID: CR-A1_M15 | Mattress ID: #A046 | Active USBL: Survey Pole
Survey Date/Time: 2026-11-04 14:34 | Deliverable Base: 20261104_Mattress_CR-A1_M15_A046
Chart Scales @ A3: Panel 1 (1:250) | Panel 2 (1:75) | Panel 3 (1:75/1:15) | Panel 4 (1:50/1:15)

## EXECUTIVE VERDICT

- Overall Status: VERIFIED - WITHIN DESIGN & GAP TOLERANCE
- Identification & Sensor: Mattress #A046 at CR-A1_M15 (CR-A1_L1) tracked via Survey Pole.
- Centerline Offset (dCL): 0.000 m vs. Tolerance +/-0.50 m (PASS) [dAlong: +0.000 m, dCross: -0.000 m].
- Heading Alignment: Design South Brg 195.854 deg vs. As-Laid 195.861 deg (dBrg: +0.007 deg).
- ROV Corner Gaps: 0.00m | Max Gap 0.00 m vs. Tolerance <=0.20 m (PASS).

## 1. GEODETIC, 4-CORNER ROV & GAP COMPARISON TABLE

| Parameter | Design Baseline | QA/QC USBL Fix | As-Laid 4-Corner ROV | Delta / Status |
|---|---|---|---|---|
| Easting E (m, TWD97) | 194541.443 | 194541.443 | 194541.443 | +0.000 m |
| Northing N (m, TWD97) | 2702613.394 | 2702613.394 | 2702613.394 | +0.000 m |
| Depth Z (m, TWVD2001) | Seabed + 0.30 m | - | -18.20 | Verified WROV |
| South Bearing (deg) | 195.854 deg | 195.861 deg | 195.861 deg | +0.007 deg |
| North Azimuth (deg) | 15.854 deg | - | 15.861 deg | +0.007 deg |
| Radial Offset dCL (m) | 0.000 m (Tol +/-0.50m) | - | 0.000 m | PASS (88.0% in box) |
| ROV Corner Gaps (m) | Tol <=0.20 m | - | 0.00m | PASS |

## 2. KEY SURVEY FINDINGS (4-CORNER ROV COORDINATES & ENVIRONMENT)

- C1 (NW): E 194540.820 m, N 2702616.690 m, Z -18.20 m | ROV Gap NW: N/A
- C2 (NE): E 194543.706 m, N 2702615.870 m, Z -18.20 m | ROV Gap NE: N/A
- C3 (SE): E 194542.066 m, N 2702610.098 m, Z -18.20 m | ROV Gap SE: N/A
- C4 (SW): E 194539.180 m, N 2702610.918 m, Z -18.20 m | ROV Gap SW: N/A
- Edge Clearances: NNE +0.00m (Nbr: None), SSW +0.00m (Nbr: None), ESE +0.00m (Nbr: None), WNW +0.00m (Nbr: No ne).
- Environment: Wind: Recorded in Log | Current: Recorded in Log.

## 3. OPERATIONAL CRANE / ROV NUDGE RECOMMENDATION

- Required Vector to Design Center: Shift 0.00 m toward N (Azimuth 0.0 deg).
- Cartesian Adjustments: dE = +0.000 m, dN = -0.000 m.

## 4. SURVEYOR REMARKS

- Mattress installation verified via WROV 4-corner fix and USBL telemetry.